

Module 4 — Read 1

Study design, two-group inference, and comparing more than two groups

Module 4, Read 1: Designing and analyzing fair comparisons

This reading is your self-paced textbook for Module 4. It has three sections. Work one section at a time and use the Check yourself box at the end of each before [Quiz A](#).

Whenever I see a difference between the groups in the data, I first ask which of three explanations could produce it: it could be by chance, it could be a real causal effect, or it could be confounding. A randomized controlled trial removes confounding as an explanation (random assignment balances other variables), and a small p-value lets me set aside the by-chance explanation. What is left is the causal explanation.

- Section 1, Study design (including confounders). How groups form, the three explanations for a difference, confounding, DAGs, and what randomization buys you.
- Section 2, Two-group tests and confidence intervals. Pick and run the right test for a two-group comparison, and interpret the p-value and 95 percent confidence interval.
- Section 3, Comparing more than two groups. The same logic extends to three or more groups: ANOVA (via `lm`) for a numerical response, and a chi-square test of independence for a binary or categorical response. Project 4 stays a two-group comparison, so this section is for awareness and your methods map.

Learning outcomes

After this reading you should be able to:

Section 1: Study design (including confounders)

- Identify the explanatory variable and the response variable in a causal question.
- Identify randomized controlled trials (RCTs) and observational study designs when two or more groups are being compared.
- Explain how units end up in comparison groups in an RCT versus an observational study.
- List the three explanations for an observed difference in an observational study: by chance, causation, confounding.
- List the two explanations for an observed difference in an RCT: by chance or causation.
- Define a confounder in words and in a DAG, and use subclassification to compare groups within levels of a measured confounder.
- Distinguish random assignment (controls confounding) from random sampling (controls bias when generalizing).

Section 2: Two-group tests and confidence intervals

- Use variable types to pick `t.test()` for two independent numerical groups or `prop.test()` / `chisq.test()` for binary comparisons, and name the conditions that match the [methods map](#).
- Identify the parameter of interest: a difference in means for a numerical response, or a difference in proportions for a binary response.
- Interpret Welch two-sample t-test output (hypotheses, p-value, 95 percent CI) in context.
- Interpret two sample proportions test/ χ^2 test output (hypotheses, p-value, 95 percent CI) in context.
- Recognize paired or matched data and use the paired t-test (numerical response) or McNemar's test (binary categorical response).

Section 3: Comparing more than two groups

- Use one-way ANOVA (`lm()` then `anova()`) for a numerical response across three or more groups.

- Use a chi-square test of independence (`chisq.test()`) for a binary or categorical response across three or more groups, and distinguish it from a goodness-of-fit test.

Section 1: Study design

Project 4 connection

Project 4 asks you to design a fair comparison between two groups. This section gives you the vocabulary: EV, RV, RCT vs observational, confounding, and what a p-value can and cannot tell you about causation.

Explanatory and response variables

A causal question asks about the effect of one thing on another, and it has two roles:

- Response variable (RV): the effect or outcome you want to compare (also called the outcome or dependent variable).
- Explanatory variable (EV): the potential cause or group label you compare (also called the treatment or exposure).

Some research questions, with the two variables identified:

- Does a tutoring program raise test scores? EV: tutoring (yes/no); RV: test score.
- Does a new drug lower blood pressure more than a placebo? EV: drug vs placebo; RV: blood pressure.
- Are remote workers more productive than in-office workers? EV: remote vs in-office; RV: a productivity measure.

When you write your own question, define each variable precisely: say how it is measured and any cutoffs (pass/fail, hours of sleep, income in dollars). Vague labels make design and analysis untrustworthy.

A story: which university teaches calculus better?

Suppose a state has two polytechnic universities, Timber Ridge and Wild Horse, and the Chancellor's Office wants to know which one does a better job teaching calculus to incoming freshmen.

*This example is a modification of a story in Pearl, Judea, and Dana Mackenzie, *The Book of Why: The New Science of Cause and Effect* (Basic Books, 2018).*

Each school has a calculus class of 200 students. Here are the crude pass rates:

	Passed calculus
Timber Ridge	150/200 (75%)
Wild Horse	154/200 (77%)

At first glance Wild Horse looks a little better, 154 passing versus 150. Before reading on, ask yourself: what could make these two numbers differ?

Maybe the difference in number passing is just **by chance**? This is only out of 200 students at each university.

Or maybe Wild Horse does a slightly better job and the different in passing rates reflects this **causation**.

Or maybe this is not a fair comparison: what if the backgrounds of the student populations at the two universities are different?

Suppose we also learn how many students in each school had already taken **AP Calculus** in high school, and we split the table by that:

	AP calculus	No AP calculus	Combined
Timber Ridge	45/50 (90%)	105/150 (70%)	150/200 (75%)
Wild Horse	136/170 (80%)	18/30 (60%)	154/200 (77%)

Each cell shows the number who passed out of the number of students in that cell. Read down each column, comparing the two schools among students with the same preparation:

- Among students without AP calculus, Timber Ridge passes more (70% vs 60%).
- Among students with AP calculus, Timber Ridge again passes more (90% vs 80%).

So within each level of preparation Timber Ridge actually does better, even though its crude rate looked worse. The comparison flipped once we accounted for AP background (a reversal known as Simpson's paradox).

Why did it flip? Wild Horse enrolled far more students who had already taken AP calculus (170 of 200 versus 50 of 200), and AP students pass at higher rates no matter which school they attend. So Wild Horse's higher crude rate came partly from starting with better-prepared students, not from better teaching.

That leaves three possible reasons the crude numbers differ at all:

1. the gap is just random variation,
2. one school really does teach calculus better, or
3. the two student bodies differ in something, here AP background, that affects passing on its own.

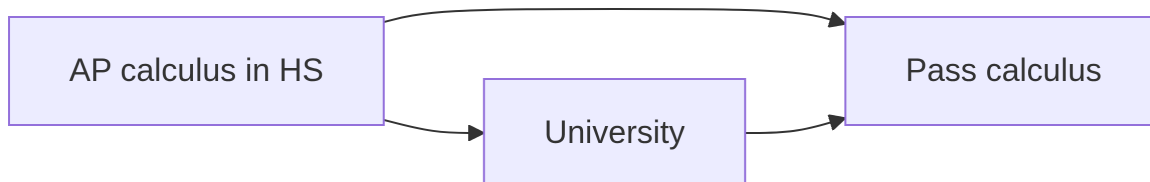
That third variable, AP background, is an example of a **confounder**: it influences both which school a student attends and whether they pass. A directed acyclic graph (DAG) is an easy way to spot one. Draw an arrow from each variable to whatever it influences; a confounder is **any variable with arrows pointing into both the explanatory variable and the response**.

flowchart LR

A[AP calculus in HS] --> U[University]

A --> P[Pass calculus]

U --> P



One clean way to get rid of a confounder is to make the groups have the same mix of it. If we could randomly assign incoming students to the two universities, AP background, and everything else, would end up balanced across the schools on average, and a crude comparison would be fair. But we cannot ethically or practically assign students to a university, so we are stuck comparing groups that formed on their own and must deal with AP background after the fact (for example by comparing within the AP and no-AP groups). The next part puts names to these two situations.

Formal definitions

With the story in mind, here is the vocabulary.

- Observational units: the entities that each get a value for every variable, usually one row of your dataset (here, one student).
- Explanatory variable (EV) and response variable (RV): as above, the group label or cause (which university) and the measured effect (passed calculus or not).
- Confounder (for the effect of an explanatory variable X on a response Y): a variable that influences both X and Y . In a DAG it has arrows pointing into both EV and RV. AP background confounds university and passing.
- Randomized controlled trial (RCT): a study in which the researcher randomly assigns units to the groups. Random assignment balances other variables, measured and unmeasured, across the groups on average, which removes confounding as an explanation.
- Observational study: a study in which group membership is not assigned by the researcher; the groups form on their own (students choose a university). Confounding is possible and must be reasoned about and adjusted for.

Internal validity is how well a study rules out alternative explanations for a difference between its own groups. Because random assignment removes confounding, an RCT has high internal validity; an observational study has to earn it through careful design and analysis.

More examples, and a caution about what is not a confounder:

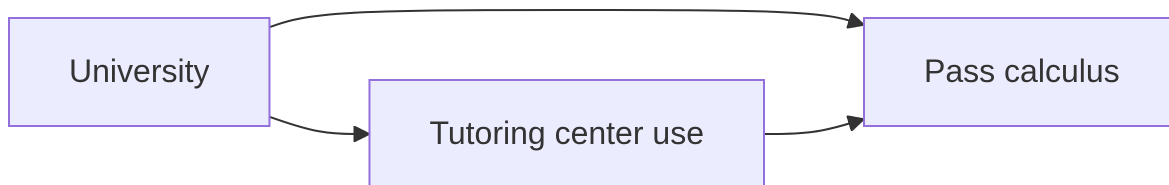
- Nightlight and myopia: parental myopia or genetics may confound nightlight use and child myopia (they could drive both).
- Tutoring that the university provides is not a confounder. It is downstream of university (university to tutoring to passing), so it lies on the causal path you care about, not on a backdoor path. Adjusting for it would remove part of the very effect you are trying to measure.

flowchart LR

U2[University] --> T[Tutoring center use]

U2 --> P2[Pass calculus]

T --> P2



A backdoor path is a non-causal route connecting EV and RV through a common cause, in the story $U \leftarrow A \rightarrow P$ (university U , AP background A , passing P). Comparing schools within the AP and no-AP groups (subclassification, also called stratification) blocks that path. You can trace backdoor paths by hand or use [DAGitty](#) to suggest adjustment sets under an assumed graph. Always ask whether your DAG includes every important common cause; unmeasured confounders (motivation, SES) can remain, which is why an RCT, when feasible, is so valuable.

Putting it together, whenever two groups differ there are three explanations to weigh:

1. By chance. Random variation alone can produce a gap, even when the groups are really the same.
2. Causation. The explanatory variable really does change the response.
3. Confounding. A third variable influences both the explanatory and response variables and distorts the comparison.

A hypothesis test and its p-value (Section 2) address the by-chance explanation only. An RCT removes confounding by design, leaving just chance and causation, so there a small p-value points to causation. An observational study leaves all three on the table, so you reason about confounding with DAGs and address measured confounders with subclassification.

Seeing confounding in data (the three arrows)

A DAG with a confounder has three arrows: confounder to group, confounder to outcome, and group to outcome. Each arrow shows up as an association you can plot and summarize with Module 3 tools. Below we simulate a treatment comparison where age group is a confounder, then look at all three arrows. Older patients are sicker (lower outcome) and, in the observational data, are more likely to receive Treatment B.

```

if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}
library(tidyverse)

set.seed(2024)
N <- 300
# Confounder: age group (younger / older), about half and half
  
```

```

age_group <- sample(c("younger", "older"), N, replace = TRUE, prob = c(0.5, 0.5))

# Arrow 1 (confounder -> group): older patients are more likely to get Treatment B
p_B <- ifelse(age_group == "older", 0.75, 0.25)
treatment <- ifelse(rbinom(N, 1, p_B) == 1, "B", "A")

# Arrow 2 (confounder -> outcome): older patients have lower outcomes.
# Arrow 3 (group -> outcome): Treatment B has a small TRUE benefit of +3.
outcome <- 70 - 12 * (age_group == "older") + 3 * (treatment == "B") + rnorm(N, 0, 6)

conf <- tibble(
  age_group = factor(age_group, levels = c("younger", "older")),
  treatment = factor(treatment, levels = c("A", "B")),
  outcome = outcome
)
glimpse(conf)

```

```

Rows: 300
Columns: 3
$ age_group <fct> younger, older, younger, younger, older, younger, older, old~
$ treatment <fct> A, B, A, A, A, A, A, B, A, A, A, A, A, A, B, A, B, B, B, B, ~
$ outcome <dbl> 74.26722, 57.20574, 68.09254, 76.40501, 67.55353, 72.69615, ~

```

Arrow 1, confounder to group. The age mix is different in the two treatment groups, so the groups are not comparable at baseline.

```
conf |> count(treatment, age_group)
```

```

# A tibble: 4 x 3
  treatment age_group     n
  <fct>      <fct>    <int>
1 A        younger   101
2 A        older     47
3 B        younger   41
4 B        older    111

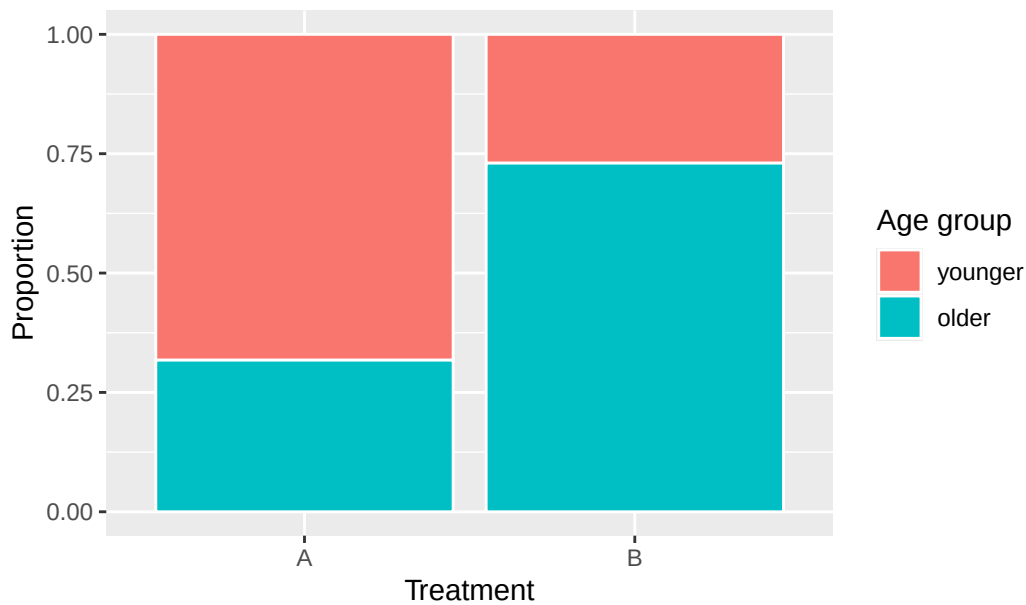
```

```

conf |>
  count(treatment, age_group) |>
  ggplot(aes(x = treatment, y = n, fill = age_group)) +
  geom_col(position = "fill", color = "white") +
  labs(title = "Arrow 1: age mix differs by treatment group (confounder to group)",
       x = "Treatment", y = "Proportion", fill = "Age group")

```

Arrow 1: age mix differs by treatment group (confounder to grc)



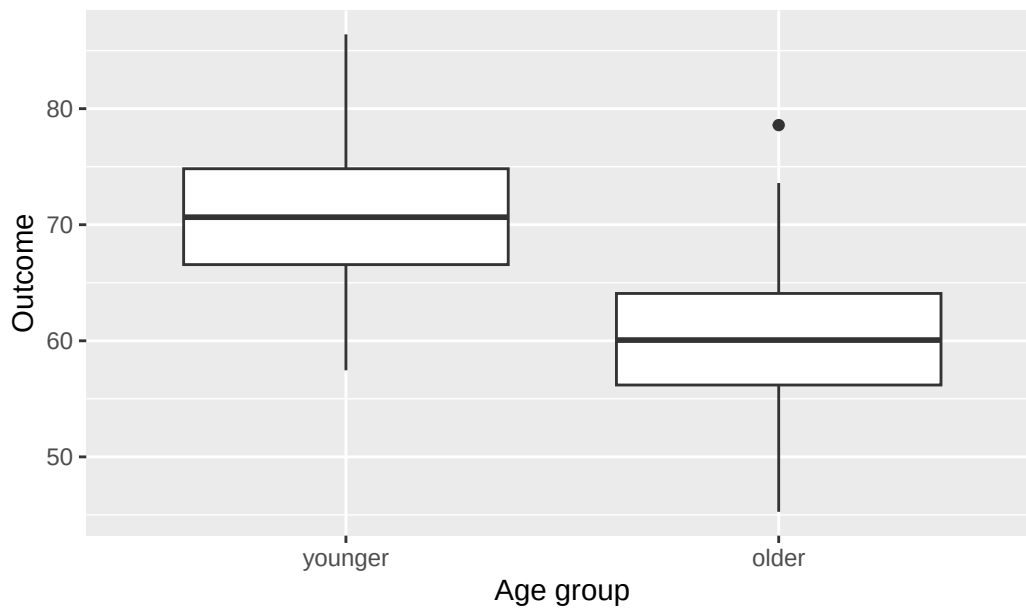
Arrow 2, confounder to outcome. Ignoring treatment, older patients have lower outcomes.

```
conf |>
  group_by(age_group) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

```
# A tibble: 2 x 4
  age_group      n mean_outcome sd_outcome
  <fct>      <int>      <dbl>    <dbl>
1 younger     142        70.7      6.15
2 older      158        60.4      5.88
```

```
ggplot(conf, aes(x = age_group, y = outcome)) +
  geom_boxplot() +
  labs(title = "Arrow 2: outcome depends on age group (confounder to outcome)",
       x = "Age group", y = "Outcome")
```

Arrow 2: outcome depends on age group (confounder to outcome)



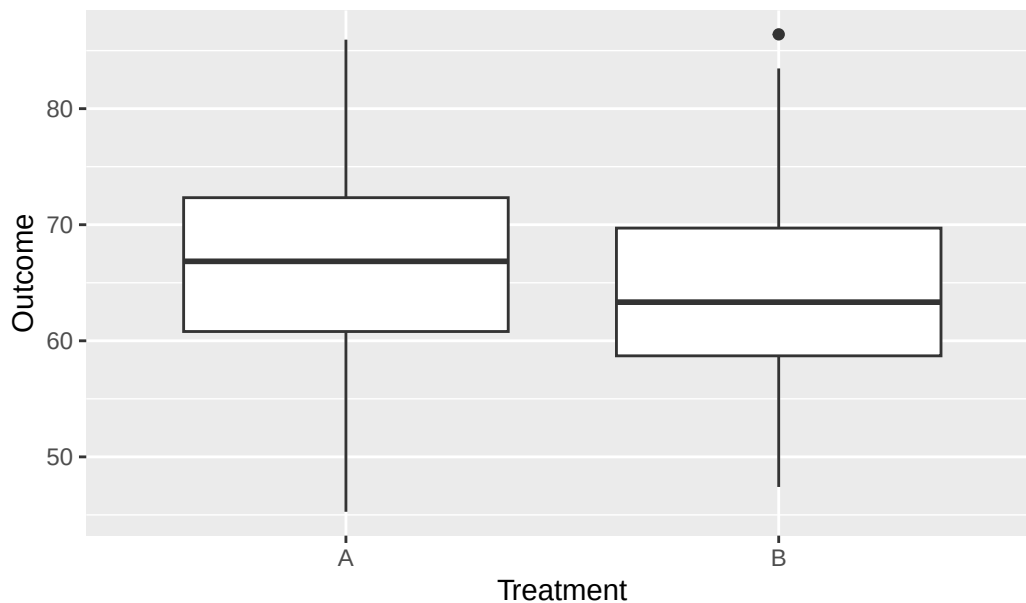
Arrow 3, the crude group-to-outcome comparison. Because older (sicker) patients were more likely to get B, the crude comparison is biased against B, even though B truly helps a little.

```
conf |>
  group_by(treatment) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

```
# A tibble: 2 x 4
  treatment      n mean_outcome sd_outcome
  <fct>      <int>      <dbl>    <dbl>
1 A          148        66.3      8.25
2 B          152        64.3      7.48
```

```
ggplot(conf, aes(x = treatment, y = outcome)) +
  geom_boxplot() +
  labs(title = "Arrow 3: crude outcome by treatment (confounded comparison)",
       x = "Treatment", y = "Outcome")
```


Arrow 3: crude outcome by treatment (confounded comparison)



```
t.test(outcome ~ treatment, data = conf, var.equal = FALSE)
```

Welch Two Sample t-test

```
data: outcome by treatment
t = 2.2036, df = 293.41, p-value = 0.02833
alternative hypothesis: true difference in means between group A and group B is not equal to 0
95 percent confidence interval:
 0.2143178 3.7964894
sample estimates:
mean in group A mean in group B
    66.27307      64.26767
```

The crude p-value can be small while pointing the wrong way: B looks worse than A because of the age imbalance, not because B is harmful. The test removed the by-chance explanation, but confounding remains.

Remove one arrow: randomize the treatment

Now keep the same people and the same age-to-outcome arrow, but assign treatment at random. Random assignment erases arrow 1 (the confounder no longer drives group membership), so age is balanced across groups and the crude comparison becomes fair.

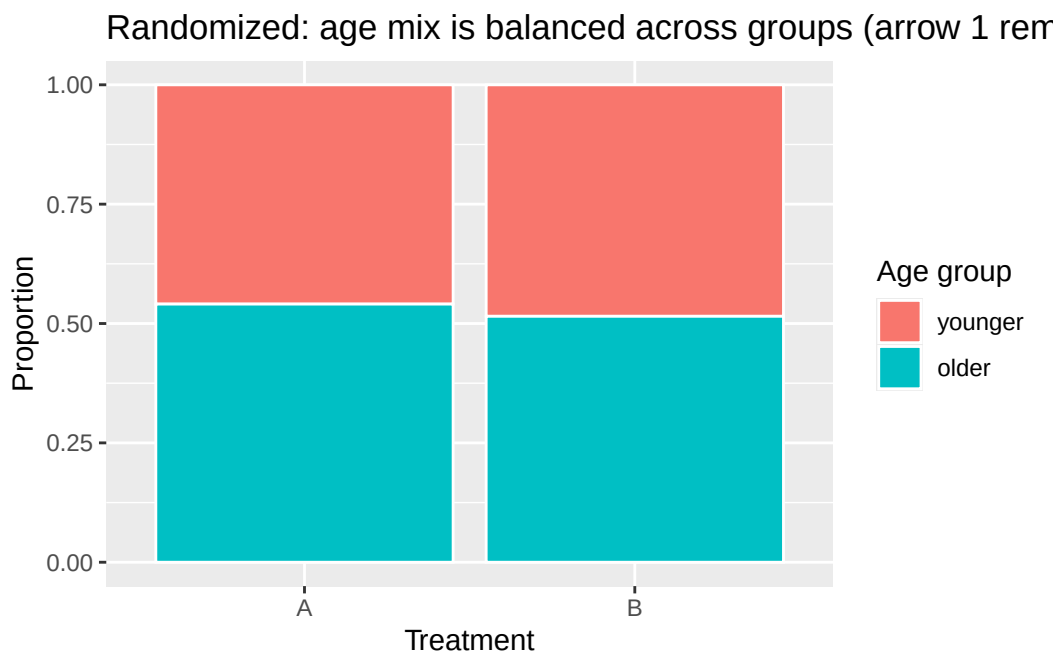
```
set.seed(2025)
treatment_rct <- sample(c("A", "B"), N, replace = TRUE) # random assignment, ignores age
outcome_rct <- 70 - 12 * (age_group == "older") + 3 * (treatment_rct == "B") + rnorm(N, 0, 6)

rct <- tibble(
  age_group = factor(age_group, levels = c("younger", "older")),
  treatment = factor(treatment_rct, levels = c("A", "B")),
  outcome = outcome_rct
)

# Arrow 1 is now gone: age mix is about equal in both groups
rct |> count(treatment, age_group)
```

```
# A tibble: 4 x 3
  treatment age_group    n
  <fct>      <fct>    <int>
1 A        younger    62
2 A        older     73
3 B        younger    80
4 B        older     85
```

```
rct |>
  count(treatment, age_group) |>
  ggplot(aes(x = treatment, y = n, fill = age_group)) +
  geom_col(position = "fill", color = "white") +
  labs(title = "Randomized: age mix is balanced across groups (arrow 1 removed)",
       x = "Treatment", y = "Proportion", fill = "Age group")
```



```
# Crude comparison now recovers the true +3 benefit of B
rct |>
  group_by(treatment) |>
  summarize(n = n(), mean_outcome = mean(outcome), sd_outcome = sd(outcome))
```

```
# A tibble: 2 x 4
  treatment    n mean_outcome sd_outcome
  <fct>      <int>      <dbl>    <dbl>
1 A         135      63.8      7.89
2 B         165      67.1      8.37
```

```
t.test(outcome ~ treatment, data = rct, var.equal = FALSE)
```

Welch Two Sample t-test

data: outcome by treatment

t = -3.5367, df = 292.03, p-value = 0.0004711

alternative hypothesis: true difference in means between group A and group B is not equal to 0
95 percent confidence interval:

-5.181136 -1.476314

sample estimates:

mean in group A	mean in group B
63.75346	67.08218

With arrow 1 removed, arrow 2 (age to outcome) still exists, but age is spread evenly across the groups, so it no longer biases the comparison. The crude difference in means now estimates the real effect of treatment, and a small p-value supports a causal conclusion.

Random assignment is not random sampling

A final caution: do not confuse random assignment with random sampling.

- Random assignment decides which group each unit goes into within your sample. It balances confounders so the comparison between groups is fair; it is what makes an RCT work and supports a causal claim within the study.
- Random sampling decides which individuals are drawn from a larger population. It reduces bias when generalizing your results back to that population.

A study can use one, both, or neither. Random assignment supports causal claims inside the study; random sampling supports generalizing beyond it. When an RCT is impossible or unethical (you cannot randomly assign people to smoke for 20 years), you are back to an observational design and must reason about confounding with DAGs and subclassification.

Check yourself, Section 1 (one per key term, plus scenarios)

Key terms

1. Explanatory variable (EV): define it.
2. Response variable (RV): define it.
3. Observational study: define it.
4. Randomized controlled trial (RCT): define it and give its key feature.
5. Random assignment: what does it accomplish?
6. Random sampling: what does it accomplish, and how is it different from random assignment?
7. By chance: what does this explanation mean for a difference between two groups?
8. Causation: what would have to be true for this explanation to be the right one?
9. Confounding: define it as an explanation for a group difference.
10. Confounder: define it and name the two arrows that define it in a DAG.
11. DAG: what do nodes and directed edges represent?
12. Backdoor path: what is it, and how do you block it?
13. Subclassification (stratification): what is it, and what is its limitation?
14. Internal validity: define it.
15. Simpson's paradox: describe what happened in the university example.

Scenarios

16. A study compares smokers and non-smokers on lung cancer rates. RCT or observational? Why?
17. In the three-arrows simulation, why did the crude treatment comparison point the wrong way before randomization?
18. After randomizing the treatment, age still affects the outcome. Why does it no longer bias the comparison?

Answers, Section 1

1. The variable thought to cause or predict the response (treatment, exposure, grouping factor).
2. The variable we measure to see whether it changes with the EV.
3. Researchers record data as-is; group membership is self-selected.
4. An experiment with random assignment of units to groups.
5. Balances measured and unmeasured confounders across groups on average.
6. Random sampling selects individuals from a population (reduces generalization bias); random assignment splits a sample into groups (controls confounding).
7. Random variation alone could produce the gap even if groups are equal.
8. The EV truly changes the RV, with chance and confounding ruled out.
9. A third variable produces or distorts the difference.
10. A common cause of EV and RV; arrows go from the confounder to the group and to the outcome.
11. Nodes are variables; an edge from A to B means A helps cause B.
12. A non-causal EV-to-RV path through a common cause; block it by subclassifying on the confounder.
13. Compare within confounder levels; only handles measured confounders.
14. How well a study rules out alternative explanations within its groups.
15. Crude rates favored Wild Horse, but Timber Ridge was higher within each AP level.
16. Observational; smoking cannot be randomly assigned.
17. Older patients were more likely to get B, so the crude difference was confounded by age.
18. Randomization balances age across groups, so it no longer biases the comparison.

Section 2: Two-group tests and confidence intervals

Design (Section 1) decides whether a difference can be called causal. This section handles the by-chance explanation. A hypothesis test asks: if the two groups were really the same, how surprising is a difference this large? The p-value answers that. A 95 percent confidence interval estimates how big the difference is, with an interval of plausible values.

The first decision is the parameter of interest, and it follows from the response variable type:

- Numerical response (a measured number per unit): the parameter is a difference in means, $\mu_1 - \mu_2$. Summarize with group means and standard deviations; compare with a Welch two-sample t-test.
- Binary response (each unit is one of two categories): the parameter is a difference in proportions, $p_1 - p_2$. Summarize with group proportions (rates); compare with `prop.test()` or, equivalently, a chi-square test of independence.

Methods table for two groups

Response variable	Plot (Module 3)	Summaries	Test	CI	Conditions (match the methods map)
Numerical	Boxplot per group	Group n , mean, SD	<code>t.test(y ~ group, var.equal = FALSE)</code>	two-sample t-interval from same call	Two samples or random assignment; independent observations; each group large or roughly normal/symmetric

Response variable	Plot (Module 3)	Summaries	Test	CI	Conditions (match the methods map)
Binary	100% stacked bar	Group counts and proportions	<code>prop.test(tab)</code> or <code>chisq.test(tab)</code>	two-proportion CI from <code>prop.test</code>	Independent observations; all expected counts at least 5 (for the z/chi-square approximation)

For paired or matched data, use the paired versions instead (covered below).

Numerical response: Welch two-sample t-test

Scenario. In a sleep study, 30 volunteers are randomly assigned to a deprived group (4 hours) or a rested group (8 hours). The next morning each completes a reaction-time task (lower is better, in milliseconds). Because assignment was random, this is an RCT, so confounding is removed by design and we only need to address the by-chance explanation.

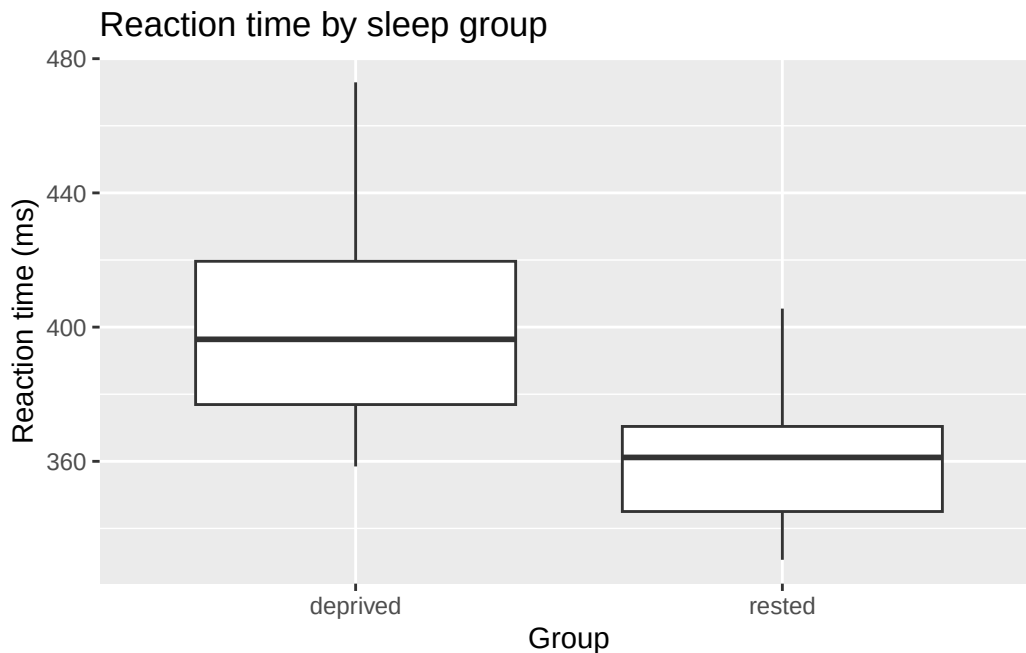
Parameter of interest: the difference in mean reaction time, $\mu_{\text{deprived}} - \mu_{\text{rested}}$.

```
library(tidyverse)
set.seed(11)
sleep_rt <- tibble(
  group = rep(c("deprived", "rested"), each = 15),
  rt = c(rnorm(15, mean = 420, sd = 40),
         rnorm(15, mean = 370, sd = 40))
)

# Module 3 summaries: n, mean, SD per group
sleep_rt |>
  group_by(group) |>
  summarize(n = n(), mean_rt = mean(rt), sd_rt = sd(rt))

# A tibble: 2 x 4
  group      n mean_rt sd_rt
  <chr> <int>   <dbl> <dbl>
1 deprived    15    403.   36.6
2 rested     15    361.   19.3

# Module 3 plot: boxplot per group
ggplot(sleep_rt, aes(x = group, y = rt)) +
  geom_boxplot() +
  labs(title = "Reaction time by sleep group", x = "Group", y = "Reaction time (ms)")
```



```
t.test(rt ~ group, data = sleep_rt, var.equal = FALSE)
```

Welch Two Sample t-test

data: rt by group

t = 3.9333, df = 21.223, p-value = 0.0007495

alternative hypothesis: true difference in means between group deprived and group rested is not equal to 0

95 percent confidence interval:

19.82771 64.25533

sample estimates:

mean in group deprived	mean in group rested
402.8768	360.8352

How to read the output:

- Hypotheses: $H_0 : \mu_{\text{deprived}} - \mu_{\text{rested}} = 0$ versus $H_a : \mu_{\text{deprived}} - \mu_{\text{rested}} \neq 0$.
- p-value: the probability of a difference at least this large if the two means were equal. A small p-value (say below 0.05) lets you set aside the by-chance explanation.
- 95 percent CI: a plausible range for $\mu_{\text{deprived}} - \mu_{\text{rested}}$. If the whole interval is above 0, deprived reaction times are slower.
- Because this is an RCT, a small p-value supports a causal reading: deprivation increased reaction time.

Conditions (from the methods map): independent observations and random assignment hold by design; with 15 per group and roughly symmetric data, the t-approximation is reasonable. Always check the boxplot for strong skew or extreme outliers first.

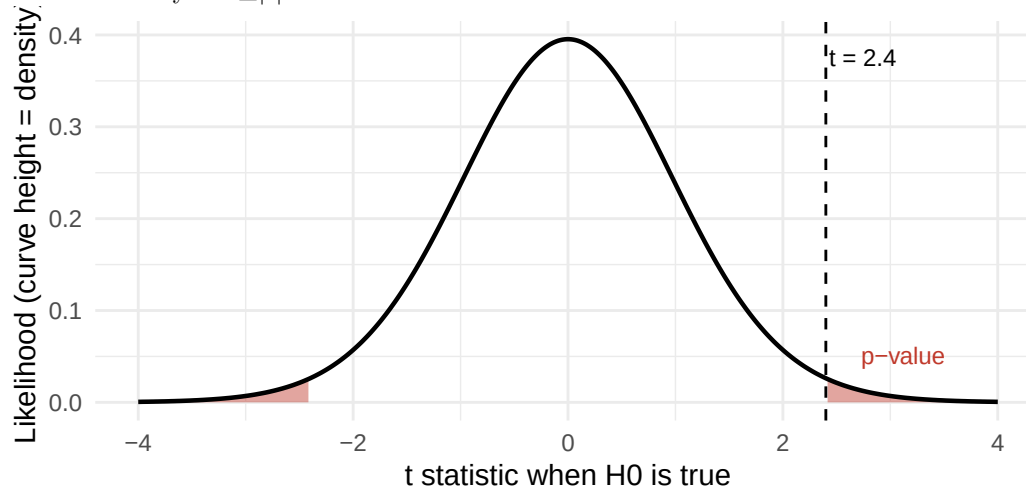
i Reference box: Welch two-sample t-test

Hypotheses: $H_0 : \mu_1 = \mu_2$ versus $H_a : \mu_1 \neq \mu_2$.

Test statistic (standardized difference in means):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0 this follows a t distribution (Welch-Satterthwaite degrees of freedom). The two-sided p-value is the shaded area beyond $\pm|t|$.



Conditions (methods map): independent observations (two independent samples or random assignment); each group large or roughly normal/symmetric.

You cannot turn this t into a p-value by hand: the p-value is the shaded area under the curve. Use R (`t.test(..., var.equal = FALSE)`); before computers, people read it from a printed t -table.

Binary response: two proportions

Scenario. A vaccine trial randomly assigns 200 volunteers to vaccine or placebo and records whether each person caught the flu that season (yes/no). The response is binary, so the parameter of interest is the difference in infection proportions, $p_{\text{vaccine}} - p_{\text{placebo}}$.

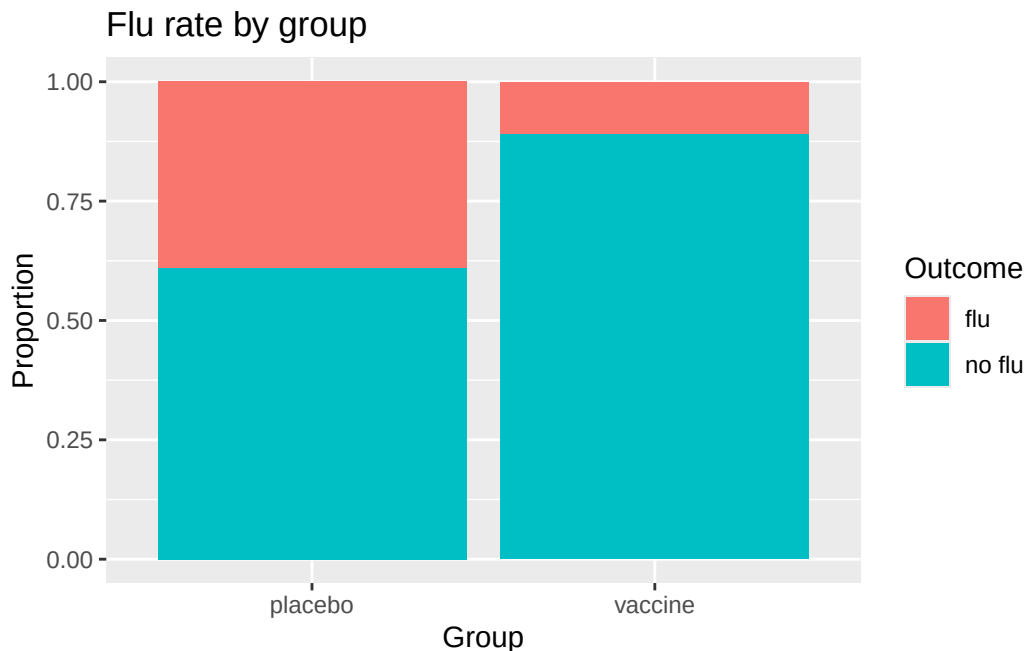
```
library(tidyverse)
set.seed(7)
vax <- tibble(
  group = rep(c("vaccine", "placebo"), each = 100),
  flu = c(rbinom(100, 1, 0.10), rbinom(100, 1, 0.30))
) |>
  mutate(flu = factor(ifelse(flu == 1, "flu", "no flu"), levels = c("flu", "no flu")))

# Module 3 summaries: counts and infection rate per group
vax |>
  group_by(group) |>
  summarize(n = n(), flu_count = sum(flu == "flu"), flu_rate = mean(flu == "flu"))
```

```
# A tibble: 2 x 4
  group      n flu_count flu_rate
  <chr> <int>   <int>   <dbl>
1 placebo  100     39     0.39
2 vaccine  100     11     0.11
```

```
# Module 3 plot: 100% stacked bar
ggplot(vax, aes(x = group, fill = flu)) +
```

```
geom_bar(position = "fill") +
labs(title = "Flu rate by group", x = "Group", y = "Proportion", fill = "Outcome")
```



```
tab <- table(vax$group, vax$flu)
tab
```

	flu	no flu
placebo	39	61
vaccine	11	89

```
prop.test(tab) # difference in proportions, with 95% CI
```

2-sample test for equality of proportions with continuity correction

```
data: tab
X-squared = 19.44, df = 1, p-value = 1.038e-05
alternative hypothesis: two.sided
95 percent confidence interval:
 0.1564235 0.4035765
sample estimates:
prop 1 prop 2
 0.39  0.11
```

```
chisq.test(tab) # equivalent test of independence for a 2x2 table
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: tab
X-squared = 19.44, df = 1, p-value = 1.038e-05
```

`prop.test()` reports the two group proportions, a p-value, and a 95 percent CI for $p_1 - p_2$. `chisq.test()` gives the same conclusion as a test of independence; use `chisq.test(tab)$expected` to confirm the condition that all expected counts are at least 5. For a 2x2 table the two tests agree.

i Inference reference box: two categorical variables

For a 2x2 comparison of a binary outcome between two groups, `prop.test(tab)` and `chisq.test(tab)` test the same null (no association between group and outcome) and reach the same conclusion. Report `prop.test()` when you also want the CI for $p_1 - p_2$. Condition: independent observations and all expected counts at least 5.

i Reference box: two-proportion test

Hypotheses: $H_0 : p_1 = p_2$ versus $H_a : p_1 \neq p_2$.

Test statistic (pooled two-proportion z):

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Under H_0 , z follows a standard normal distribution; `prop.test()` reports the equivalent $\chi^2 = z^2$. The two-sided p-value is the shaded area beyond $\pm|z|$.



Conditions (methods map): independent observations; all expected counts at least 5.

You cannot turn this z into a p-value by hand; use R (`prop.test()` / `chisq.test()`), or in olden times a printed normal (or χ^2) table.

Paired or matched comparisons

Sometimes the two measurements are linked: the same unit measured twice (before/after), or units matched into pairs. Then the observations are not independent across groups, and the independent-samples tests above do not apply. Use the paired versions, which analyze the within-pair differences.

Numerical outcome, paired t-test. Example: blood pressure for 12 patients, before and after a program. Test the mean of the before-minus-after differences.

```
library(tidyverse)
set.seed(3)
bp <- tibble(
  patient = 1:12,
  before = rnorm(12, 150, 10),
  after = rnorm(12, 142, 10)
)
```

```
bp |> summarize(mean_before = mean(before), mean_after = mean(after),
                mean_change = mean(before - after))
```

```
# A tibble: 1 x 3
  mean_before mean_after mean_change
      <dbl>      <dbl>      <dbl>
1      148.       139.        9.37
```

```
t.test(bp$before, bp$after, paired = TRUE)
```

Paired t-test

```
data: bp$before and bp$after
t = 3.298, df = 11, p-value = 0.007103
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 3.115704 15.617766
sample estimates:
mean difference
 9.366735
```

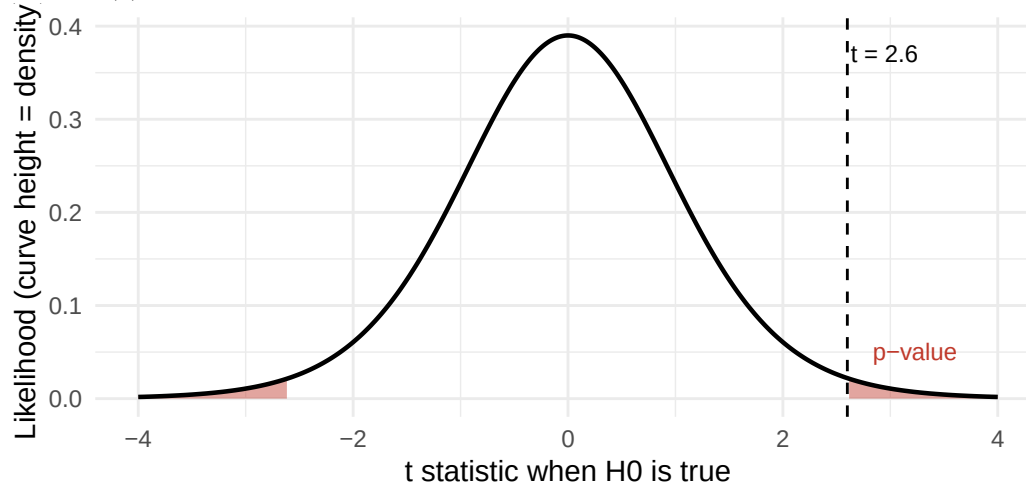
i Reference box: paired t-test

Let d_i be the within-pair differences (for example before minus after). Hypotheses: $H_0 : \mu_d = 0$ versus $H_a : \mu_d \neq 0$.

Test statistic:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Under H_0 this follows a t distribution with $n - 1$ degrees of freedom. The two-sided p-value is the shaded area beyond $\pm|t|$.



Conditions (methods map): pairs are independent of each other; the within-pair differences are roughly normal or n is large.

You cannot turn this t into a p-value by hand; use R (`t.test(x, y, paired = TRUE)`), or in olden times a printed t -table.

Binary outcome, McNemar's test. Example: 80 people rated a policy support yes/no before and after a debate. The 2x2 table cross-classifies each person's before and after answers; McNemar's test uses only the people who switched.

```
support <- matrix(c(20, 25, 10, 25),
                  nrow = 2, byrow = TRUE,
                  dimnames = list(before = c("yes", "no"),
                                  after = c("yes", "no")))

support
```

```
      after
before yes no
yes    20 25
no     10 25
```

```
mcnemar.test(support)
```

McNemar's Chi-squared test with continuity correction

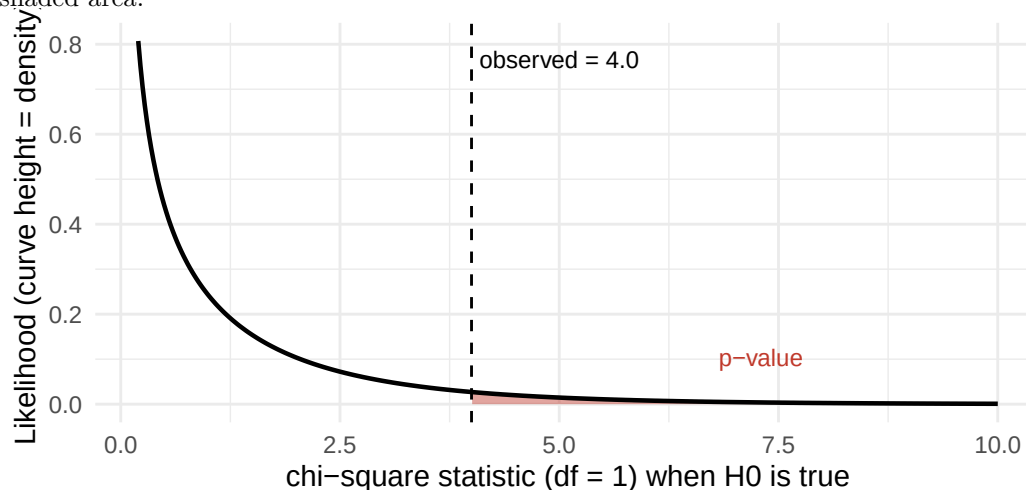
```
data: support
McNemar's chi-squared = 5.6, df = 1, p-value = 0.01796
```

i Reference box: McNemar's test

For paired binary data, let b and c be the two discordant counts (the pairs that switched, for example yes-then-no and no-then-yes). Hypotheses: H_0 : the two marginal rates are equal (no net change) versus H_a : they differ. Test statistic:

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

Under H_0 this follows a χ^2 distribution with 1 degree of freedom. Because $\chi^2 \geq 0$, the p-value is the upper-tail shaded area.



Conditions (methods map): paired binary observations; enough discordant pairs ($b + c$ not too small). You cannot turn this χ^2 into a p-value by hand; use R (`mcnemar.test()`), or in olden times a printed χ^2 table.

Independence vs paired is a design question, not a data question. Ask: was each value in group 1 produced by the same unit (or a matched partner) as a specific value in group 2? If yes, the data are paired and you use the paired test; if the two groups are separate, unrelated units, use the independent-samples test.

Ruling out chance without a formula (optional)

You can also address the by-chance explanation by simulation: shuffle the group labels many times, recompute the difference, and see how often a difference as large as the observed one appears by chance alone. The free

[Rossman-Chance applets](#) let you do this for both means and proportions, and the demo labs show the same idea in R with `set.seed()` and resampling.

Check yourself, Section 2 (key terms, scenarios, paired-or-not)

Key terms

1. By-chance explanation: what does a hypothesis test do about it?
2. p-value: state its meaning in one sentence.
3. Confidence interval (95 percent): what does it estimate, and what does it mean if it contains 0?
4. Parameter of interest: how do you decide between a difference in means and a difference in proportions?
5. Welch two-sample t-test: when do you use it, and what are its conditions?
6. Two-proportion test (`prop.test`): when do you use it, and what are its conditions?
7. Chi-square test of independence: what null does it test, and what is the expected-count condition?
8. Paired t-test: when do you use it instead of the two-sample t-test?
9. McNemar's test: when do you use it?

Choose the test/CI and state its conditions (use the methods map)

10. A clinic randomly assigns 60 patients to drug or placebo and records cholesterol (mg/dL) after 8 weeks. Identify the design, the parameter of interest, the test, and its conditions.
11. A survey records whether each of 300 adults exercises (yes/no) and whether they own a dog (yes/no), and asks if dog owners exercise more. Identify the design, parameter, test, and conditions.
12. Researchers measure each of 25 runners' resting heart rate before and after a 10-week program. Identify the design, parameter, test, and conditions.

Paired or not?

13. Group 1 is 40 men and group 2 is 40 different women; compare mean income. Paired or independent?
14. Each of 30 students takes a pre-test and a post-test; compare mean scores. Paired or independent?
15. 50 patients are each measured on the left and right arm; compare mean blood pressure. Paired or independent?

Answers, Section 2

1. It measures how surprising the difference is under the null and lets you set aside chance when small.
2. Probability of a difference at least this large if the null (no difference) is true.
3. A plausible range for the true difference; containing 0 means consistent with no difference.
4. Numerical response gives a difference in means; binary response gives a difference in proportions.
5. Two independent numerical groups; conditions: independence/random assignment, each group large or roughly symmetric.
6. Two independent groups, binary outcome; conditions: independence, expected counts at least 5.
7. Null: group and outcome independent; condition: independence and expected counts at least 5.
8. When measurements are linked (before/after or matched pairs).
9. Binary outcome measured twice on the same units (or matched pairs).
10. RCT; difference in means; Welch t-test + t-interval; independence/random assignment, each group large or roughly symmetric.
11. Observational; difference in proportions; prop.test/chisq.test; independence, expected counts at least 5; association not causation.
12. Paired; mean difference; paired t-test + t-interval; differences roughly normal or large n.
13. Independent. 14. Paired. 15. Paired.

Section 3: Comparing more than two groups

i Scope for Project 4

Project 4 stays a single two-group comparison (one binary explanatory contrast). This section is for awareness and your methods map, so you recognize the right tool when there are three or more groups.

The three-explanations logic does not change with more groups: a test and p-value handle the by-chance explanation. Only the test changes, because comparing three or more group means at once with many t-tests inflates the chance of a Type I error (false discovery).

Numerical response, three or more groups: one-way ANOVA

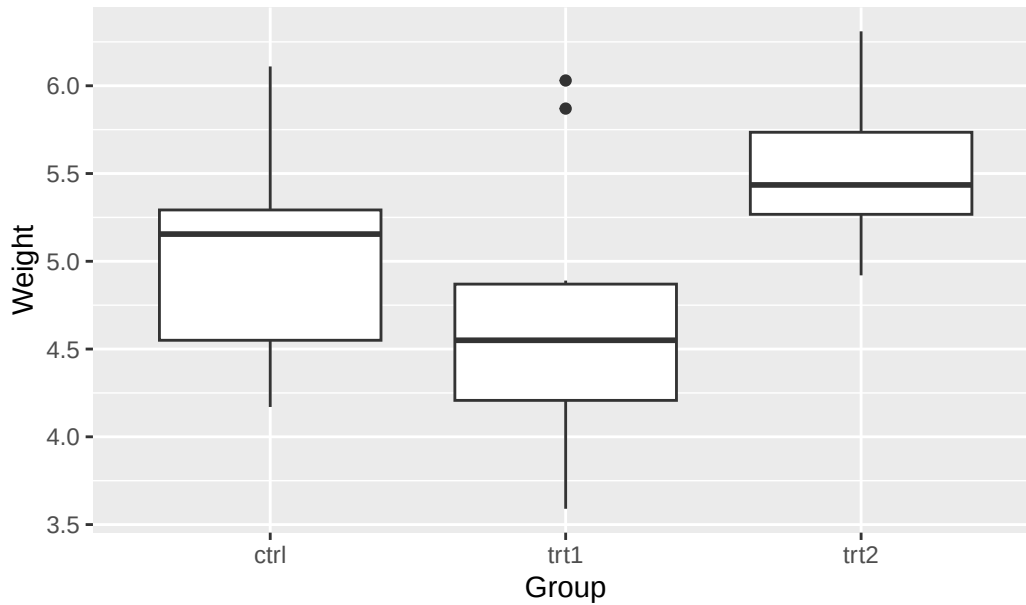
One-way ANOVA tests whether all group means are equal: $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$. Fit a linear model with `lm()` and read the overall test with `anova()`.

```
library(tidyverse)
PlantGrowth |>
  group_by(group) |>
  summarize(n = n(), mean_weight = mean(weight), sd_weight = sd(weight))
```

```
# A tibble: 3 x 4
  group      n mean_weight sd_weight
<fct> <int>      <dbl>      <dbl>
1 ctrl      10      5.03      0.583
2 trt1      10      4.66      0.794
3 trt2      10      5.53      0.443
```

```
ggplot(PlantGrowth, aes(x = group, y = weight)) +
  geom_boxplot() +
  labs(title = "Plant weight by treatment group", x = "Group", y = "Weight")
```

Plant weight by treatment group



```
fit <- lm(weight ~ group, data = PlantGrowth)
anova(fit)
```

Analysis of Variance Table

Response: weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	3.7663	1.8832	4.8461	0.01591 *
Residuals	27	10.4921	0.3886		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The `anova()` p-value tests whether any group mean differs. Conditions (from the methods map): independent observations, roughly normal within groups, and similar spread across groups. A small p-value says at least one group differs; follow-up comparisons say which.

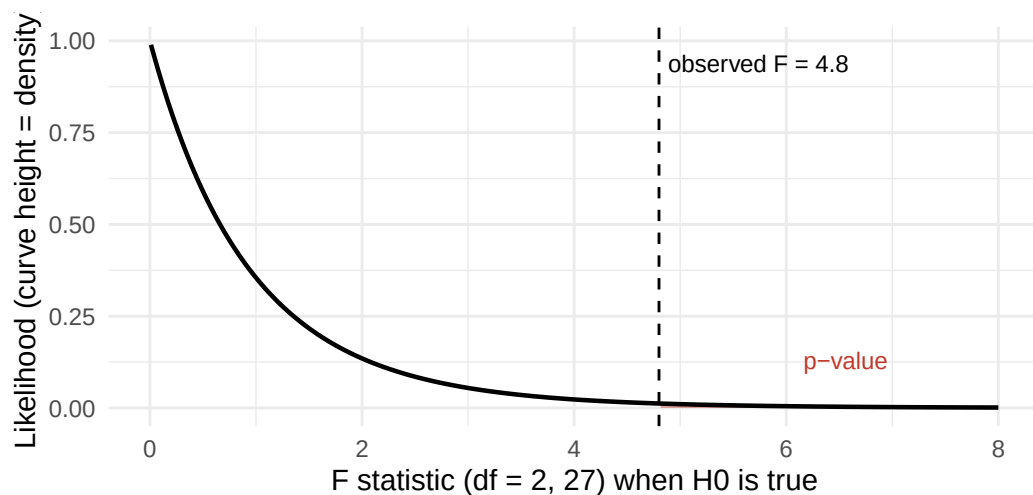
i Reference box: one-way ANOVA (F-test)

Hypotheses: $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ versus H_a : at least one group mean differs.

Test statistic (between-group variability over within-group variability):

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

Under H_0 this follows an F distribution with $k - 1$ and $N - k$ degrees of freedom. Because $F \geq 0$, the p-value is the upper-tail shaded area.



Conditions (methods map): independent observations; roughly normal within groups; similar spread across groups.

You cannot turn this F into a p-value by hand; use R (`lm()` then `anova()`), or in olden times a printed F table.

Binary response, three or more groups: chi-square test of independence

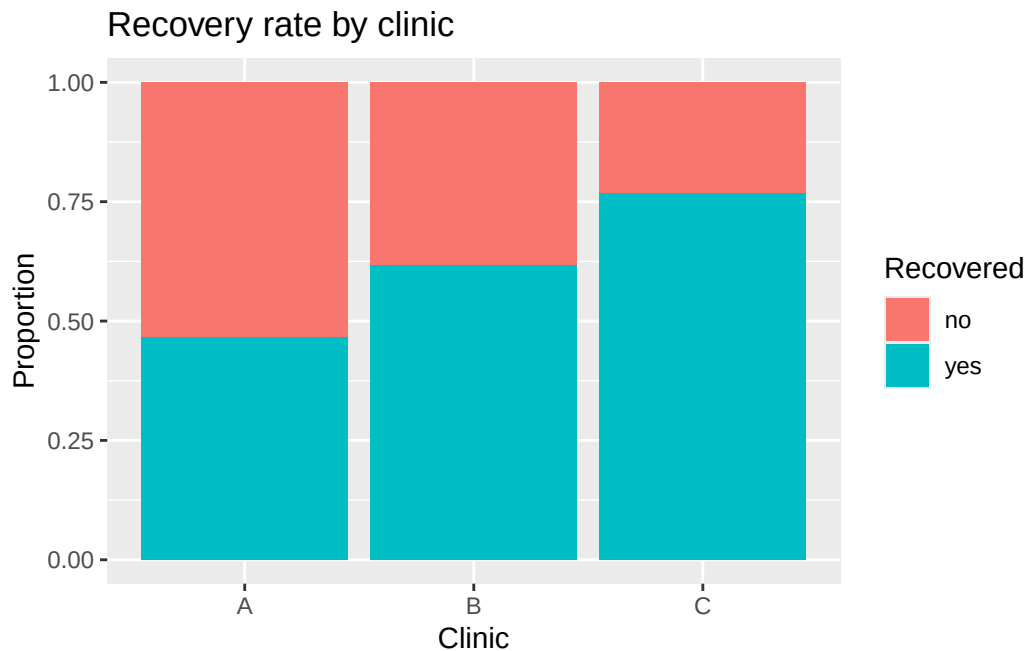
When the response is binary (or categorical) and you compare three or more groups, cross-classify group and outcome in a contingency table and run a chi-square test of independence with `chisq.test()`. It tests H_0 : the outcome rate is the same in every group (group and outcome are independent).

```
set.seed(5)
clinic <- tibble(
  clinic = factor(rep(c("A", "B", "C"), each = 60)),
  recovered = c(rbinom(60, 1, 0.5), rbinom(60, 1, 0.65), rbinom(60, 1, 0.8))
)
```

```
clinic |>
  group_by(clinic) |>
  summarize(n = n(), recovered_rate = mean(recovered))
```

```
# A tibble: 3 x 3
  clinic      n recovered_rate
  <fct> <int>         <dbl>
1 A         60         0.467
2 B         60         0.617
3 C         60         0.767
```

```
ggplot(clinic, aes(x = clinic, fill = factor(recovered, labels = c("no", "yes")))) +
  geom_bar(position = "fill") +
  labs(title = "Recovery rate by clinic", x = "Clinic", y = "Proportion", fill = "Recovered")
```



```
tab <- table(clinic$clinic, factor(clinic$recovered, labels = c("no", "yes")))
tab
```

```
  no yes
A 32 28
B 23 37
C 14 46
```

```
chisq.test(tab)$expected # all should be >= 5
```

```
  no yes
A 23 37
B 23 37
C 23 37
```

```
chisq.test(tab)
```

Pearson's Chi-squared test

```
data: tab
X-squared = 11.422, df = 2, p-value = 0.00331
```

Conditions (from the methods map): independent observations and all expected counts at least 5 (check with `chisq.test(tab)$expected`). A small p-value says the recovery rate differs across at least one clinic.

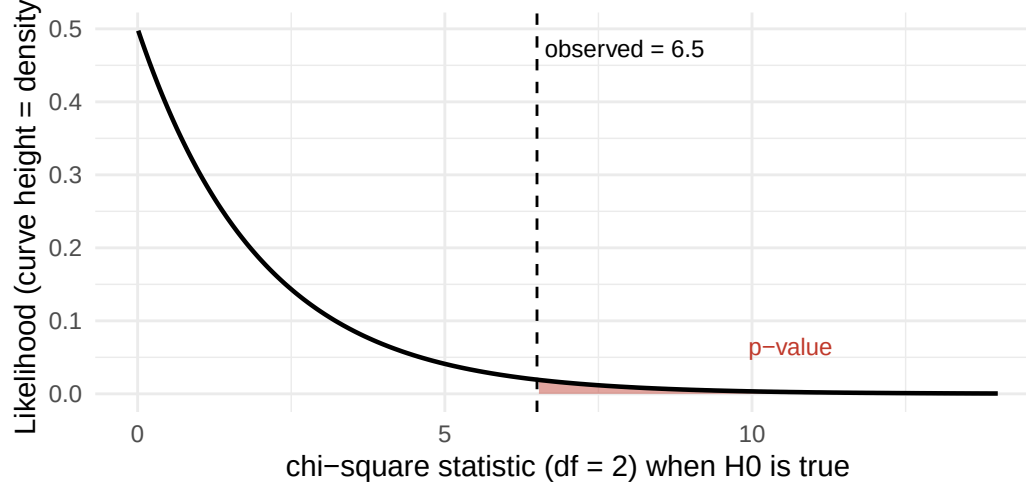
i Reference box: chi-square test of independence

Hypotheses: H_0 : group and outcome are independent (equal outcome rates across groups) versus H_a : they are associated.

Test statistic (observed vs expected counts):

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Under H_0 this follows a χ^2 distribution with $(r - 1)(c - 1)$ degrees of freedom. Because $\chi^2 \geq 0$, the p-value is the upper-tail shaded area.



Conditions (methods map): independent observations; all expected counts at least 5 (check with `chisq.test(tab)$expected`).

You cannot turn this χ^2 into a p-value by hand; use R (`chisq.test()`), or in olden times a printed χ^2 table.

Two chi-square tests, two different questions. Both use the same statistic, so it helps to keep them straight:

- Test of independence (what we use here): one sample, each unit classified by two categorical variables (group and outcome). H_0 is that the two variables are unrelated, that is, the outcome rate is the same in every group.
- Goodness-of-fit test: one categorical variable compared against a claimed set of proportions (is a die fair? do blood types match the textbook ratios?). H_0 is that the data follow that specified distribution.

When the table is just 2 groups by 2 outcomes, the chi-square test of independence tests the same null as the two-sample `prop.test()`, namely equal outcome proportions in the two groups ($p_1 = p_2$), and with `correct = FALSE` the two give the same p-value. Use `prop.test()` when you also want the confidence interval for $p_1 - p_2$; reach for `chisq.test()` once there are more than two groups (or more than two outcome categories), where a single difference in proportions no longer summarizes the comparison.

Check yourself, Section 3

1. One-way ANOVA: what null does it test, and what R code produces the overall p-value?
2. Why not just run many two-sample t-tests across several groups?
3. State the ANOVA conditions from the methods map.
4. Chi-square test of independence: what null does it test for a binary response across several groups, and how is it different from a goodness-of-fit test?
5. When does the chi-square test of independence reduce to the same null as a two-sample `prop.test()`?
6. A study compares mean recovery time across four hospitals (numerical response). Which tool, and what plot would you show first?

Answers, Section 3

1. All group means equal; `lm(y ~ group)` then `anova(fit)`.
 2. Many pairwise tests inflate the false-positive rate; ANOVA is one overall test.
 3. Independent observations, roughly normal within groups, similar spread.
 4. Categorical response across groups; `chisq.test(tab)` (expected counts at least 5); tests independence (equal rates), unlike goodness-of-fit, which checks one variable against claimed proportions.
 5. For a 2 by 2 table it tests $p_1 = p_2$, the same null as a two-sample `prop.test()` (same p-value with `correct = FALSE`).
 6. One-way ANOVA; boxplot per hospital first.
-

Where to go next

- Take [Quiz A](#) to check definitions, study-design identification, and method choice.
- Work the [demo labs](#): Demo 1 simulates data by study design and compares groups with Module 3 plots and summaries; Demo 2 runs the tests and confidence intervals from Section 2.
- Review the [methods map](#) so your test choice and stated conditions always match the course reference.
- Start [Project 4](#), keeping to a single two-group comparison.